parkflow ⇌

A member service for KL East Mall

A better arrival.
A reason to return.

Priority parking, verified at the gate.
A measured operating and investment case.

Team Hexa

Universiti Malaya · Hackathon 2026



Sources: S1: KL East Mall official parking information. Property image: official KL East Mall website. Independent Team Hexa proposal.

The team behind ParkFlow

A working prototype and a transparent investment case, built across five complementary roles.

Team Hexa

Universiti Malaya

Product, models, operations and finance. **One pilot decision.**

Five students building a member parking proposition that can be tested in a real mall.

Leow Shen En	**Product & experience** Team lead · Data Science
Pohyi Chong	**Forecasting & models** Data Science
Isaac Teo	**Analysis & simulation** Data Science
Lee Jia Shern	**Business case & finance** Actuarial Science
Choong Zhuo Lin	**AI engineering & prototype** Artificial Intelligence

A bounded pilot, with a clear investment gate

The decision is whether to validate the service at KL East before committing to a wider rollout.

Recommendation

Prove one zone. Then decide to scale.

Approve site discovery and a measured KL East pilot, with deployment funding released against a confirmed scope.

RM92k
Initial scope subtotal
Final installed cost to be confirmed

The opportunity

Make arrival part of the member benefit.

KL East has existing LPR. ParkFlow adds a proposed booking and zone-capacity service above the current payment journey.

The operating test

Protect the promise without wasting space.

Validate the deployed admission rule, real member uptake, local camera performance and the exception-handling SOP.

The investment test

Measure complete cost and incremental value.

Reconcile installed scope, central support and landlord capture before applying SDP's agreed return hurdle.

Sources: S1: KL East Mall official parking information · M1: ParkFlow shared booking economics. Budget and demand are planning inputs; no KL East field pilot or signed commercial approval is represented.

Start with the property, not a generic parking problem

Existing LPR makes integration and reuse the first design decision.



KL East Mall · proposed lighthouse site

Build on what already exists.

Published today	Cashless parking, LPR and online payment.
Proposed addition	Bookable member-zone access near the lifts.
First investigation	Verify vendor interfaces, physical boundary and zone-level counts.

Local search time, peak demand and willingness to pay require a baseline. Broad urban surveys are not a KL East demand estimate.

Sources: S1: KL East Mall official parking information. A proposed B1 Priority Zone is a concept; existing zone availability and physical feasibility are not established.

Differentiate on the operating promise

Published parking programmes establish the category; the pilot must establish ParkFlow's added value.

1 utama It's all in one · ONE CARD privileges+	**Member access is established** ONECARD publishes member preferred parking and LPR-linked access/payment.	Compare the fulfilment promise, not membership alone.
SUNWAY PYRAMID	**Premium areas and reservations exist** Sunway's FAQ covers preferred parking and EV bay reservation via ParkEasy.	Reservation is a capability to evaluate, not a uniqueness claim.
parkflow ⇌	**A pooled, auditable member service** Verified admissions, advisory forecasts and a shared record of commitments and exceptions.	Prove reliability, capacity efficiency and incremental member value.

Sources: S1: KL East Mall official parking information · S2: 1 Utama ONECARD official FAQ · S3: Sunway Pyramid official FAQ. Capabilities reflect reviewed public pages. Unmentioned capabilities are not treated as absent.

Make the service tangible

A dedicated zone by the lifts, with controlled entry and reconciled exit.



Illustrative concept only. Not a surveyed or approved KL East floor plan.

One shared zone. Any free bay.

Members reserve access to pooled capacity. The system does not assign an individual parking bay.

Boundary, arrival window, cancellation terms and assistance require operator approval.

Reserve	Verify	Park	Release
Eligible member + plate	Entitlement + capacity	Any available zone bay	Exit closes the record

A member journey that stays in one place

Booking, guidance and help should feel like a single service.

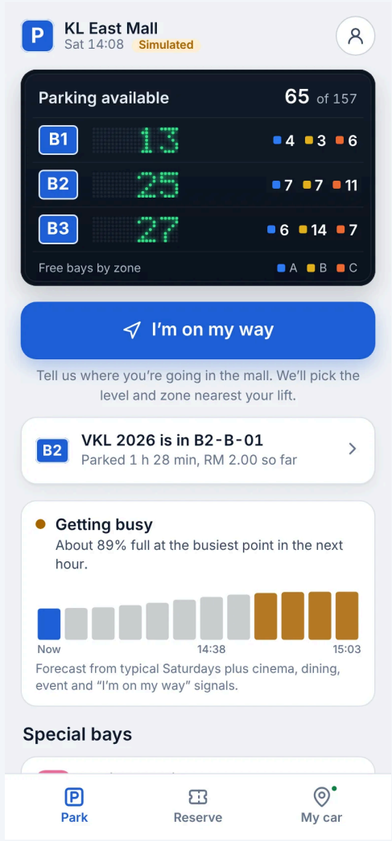
Before the journey, remove the uncertainty.

Choose See eligibility, the extra booking fee and an arrival window.

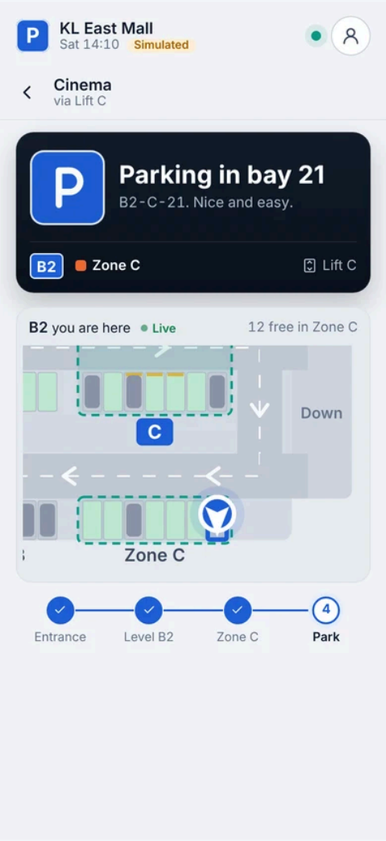
Arrive Use the registered plate and directions to the correct zone.

Resolve Reach support for late arrival, a plate mismatch or a refund.

RM4 weekday / RM5 peak is a proposed pricing test, additional to normal parking charges.



Find and reserve

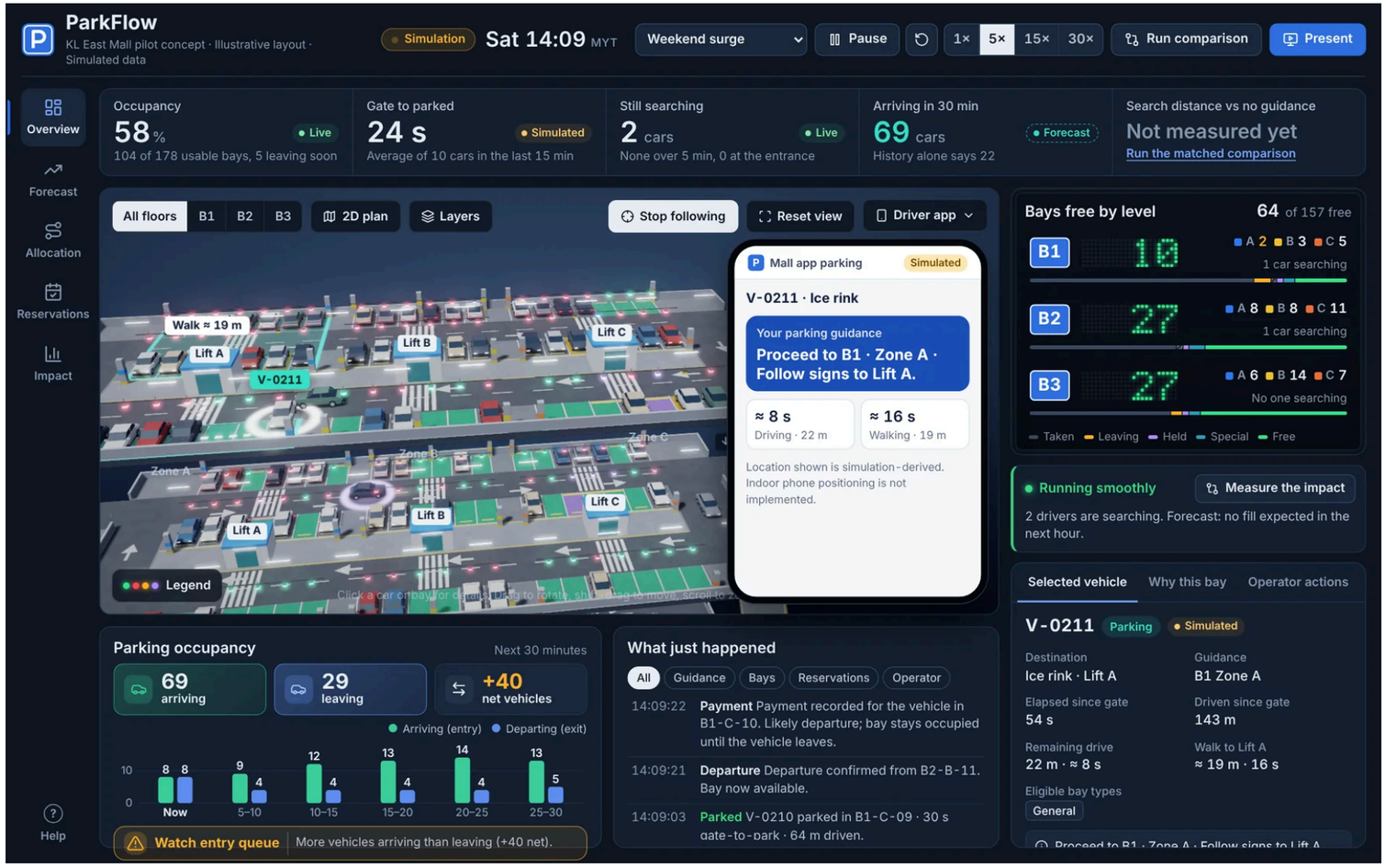


Navigate and arrive

Prototype screens use simulated data. RM4 weekday / RM5 peak fees are a proposed pricing test, separate from normal parking charges.

Give the operator one coherent view

The operational record connects entry, occupancy, commitments and exceptions.



Working interface prototype · simulated data

A record the operator can explain.

Before entry	Check promises and reconciled capacity.
During service	Monitor stale data, no-shows and exceptions.
At review	Trace every override to its reason, user and timestamp.

Assistance and emergency egress remain available during faults.

Protect the commitment at the gate

Current verified capacity and forecast availability serve different purposes.

Eight verified free bays, accounted for



Accounting illustration. Individual physical bays are not assigned.

8	–	5	–	1	=	2
verified free		outstanding promises		proposed buffer		additional promises now

Verified control protects access

Commit only against a reconciled capacity guard. Freeze new promises when counts or connectivity become uncertain.

Forecasts improve planning

Advise arrival choices and operating modes. Predicted departures alone do not create bookable capacity.

Sources: M2: ParkFlow policy simulation, recorded output. Illustrative operating rule. Buffer, promise horizon and fault handling need validation; simulation evidence uses policies with different admission logic.

Close capability gaps without duplicating the site

The public record establishes LPR; the survey must establish the additional integration and control requirements.

Capabilities to validate

Site / integration question	Why it matters
Zone-specific counts Can existing entry/exit events isolate the proposed boundary?	The capacity guard needs a reconciled zone count, not only a mall-wide payment event.
Member entitlements Can eligibility, plate and booking expiry connect to the controller?	The operator needs one valid admission decision and a traceable exception path.
Local perception Can existing CCTV resolve indoor glare, pillars and blind spots?	Buy cameras only after local coverage and false-free errors are tested.

First action: a focused site and vendor audit, before expanding the hardware scope.



Existing property, potential reuse.

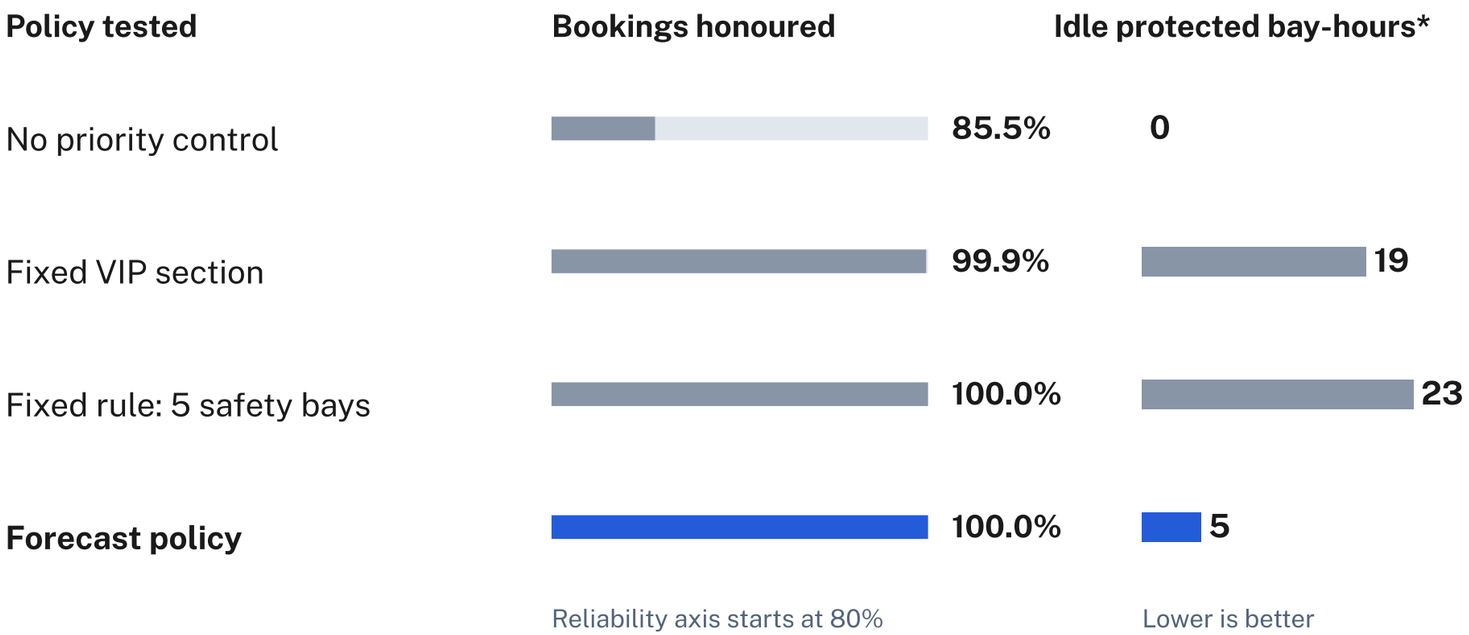
Build on the installed foundation

LPR	Published by KL East; confirm zone access and protocol.
CCTV	Survey coverage before pricing additional cameras.
Mall touchpoint	Confirm member identity, API rights and payment settlement.

Sources: S1: KL East Mall official parking information · S8: Malaysia PDP official Act/guidelines index. Proposed architecture. Neither integration permission nor incumbent vendor compatibility has been established.

Show the operating trade-off transparently

The recorded forecast policy is efficient in simulation; the deployed admission rule still needs validation.



1,000 simulated busy days · 100-bay zone · same synthetic demand.

What the model shows

A trade-off between reliability and unused protected capacity under different policies.

What the pilot must test

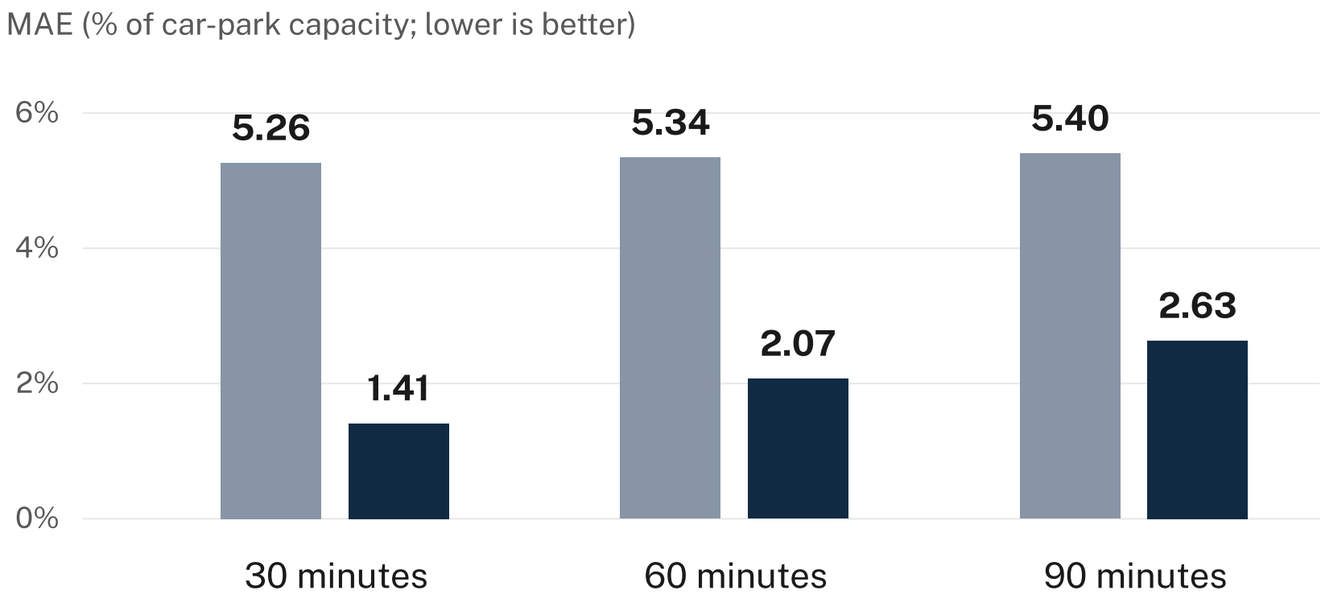
The proposed current-capacity guard differs from the forecast policy. Its efficiency is not established by this comparison.

Sources: M2: ParkFlow policy simulation, recorded output. 1,000 simulated busy days; 100-bay zone. *Free protected bay-hours only while walk-ins are refused; not all held capacity. No KL East field result.

Forecasting evidence

The ensemble beats a same-time-last-week baseline; indoor perception and KL East transfer remain separate tests.

Historical occupancy back-test



48,112 forecasts · 30 Birmingham car parks · six-week test. Lower is better.

Same time last week Ensemble

Evidence boundary

60-minute MAE: 2.07% versus 5.34% of capacity, a 61.2% relative reduction.

Camera test: 99.05% patch accuracy on 78,270 patches from held-out cameras at one outdoor Italian site.

Before deployment: test indoor lighting, pillars, occlusion, plate errors and occupancy drift at KL East.

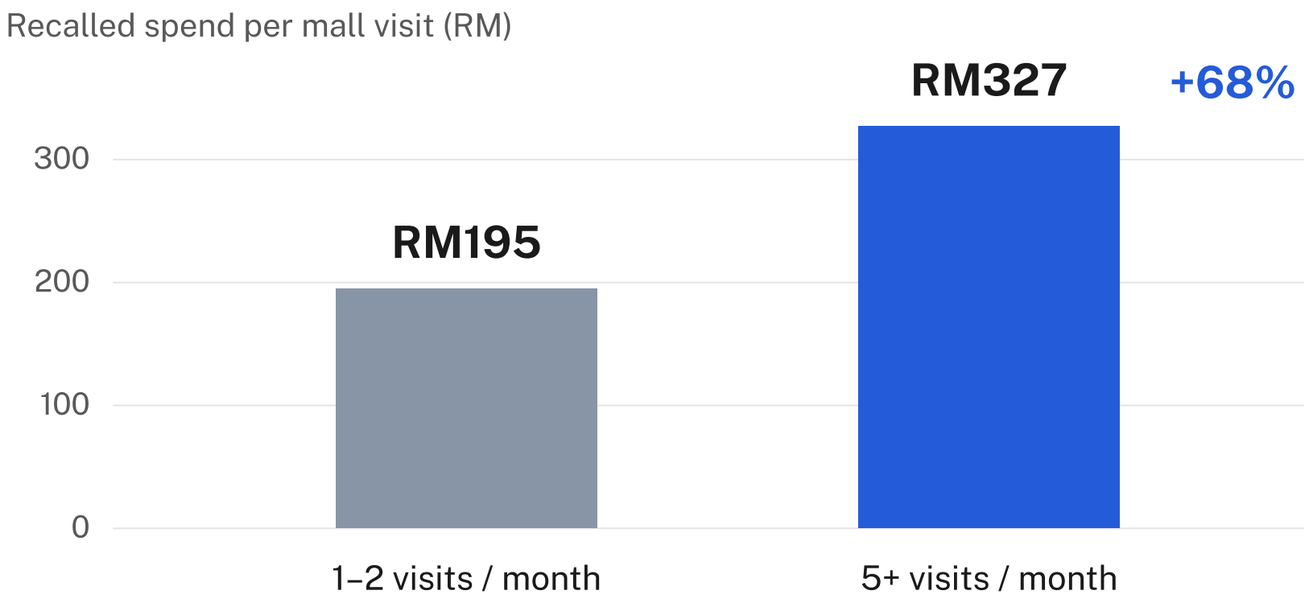
New camera views at the same location are not a new-site validation. Forecast and perception scores measure different tasks.

Sources: M3: ParkFlow ensemble back-test, recorded output · M4: ParkFlow camera benchmark, recorded output · S9: CNRPark dataset, CNR institute record · S10: UCI Parking Birmingham dataset.

Member value case

Frequent visitors report higher baskets; parking must demonstrate that it causes additional valuable visits.

Observed association



Vase.ai 2026 consumer survey; total n=500. Recalled trip spending by visit frequency.

Commercial interpretation

- Prioritize members whose repeat visits are valuable to the mall.
- Separate tenant turnover from the landlord's effective share.
- Measure incremental trips against a comparable group or baseline.
- Avoid converting the 68% basket difference directly into a parking-driven uplift.

Value bridge: additional visits × spend per visit × effective mall capture, less additional service and support costs.

Sources: S4: Vase.ai, Malaysia Shopping Mall Report 2026 (n=500). Survey association is not causal and is not a KL East estimate. Effective capture is a finance-calibrated scenario.

Pilot investment

RM92k is an initial scope subtotal; complete the installed scope and support estimate before approving funding.

Initial planning scope



Hardware RM21.5k + build RM60.0k + annual zone running RM10.5k.

Complete the approval envelope

- Survey whether existing LPR/CCTV can be reused.
- Price physical zone separation, barriers, controller and intercom interfaces.
- Confirm cabling, power, network and commissioning.
- Add central support, tax treatment and contingency.
- Define warranty, maintenance and ownership at handover.

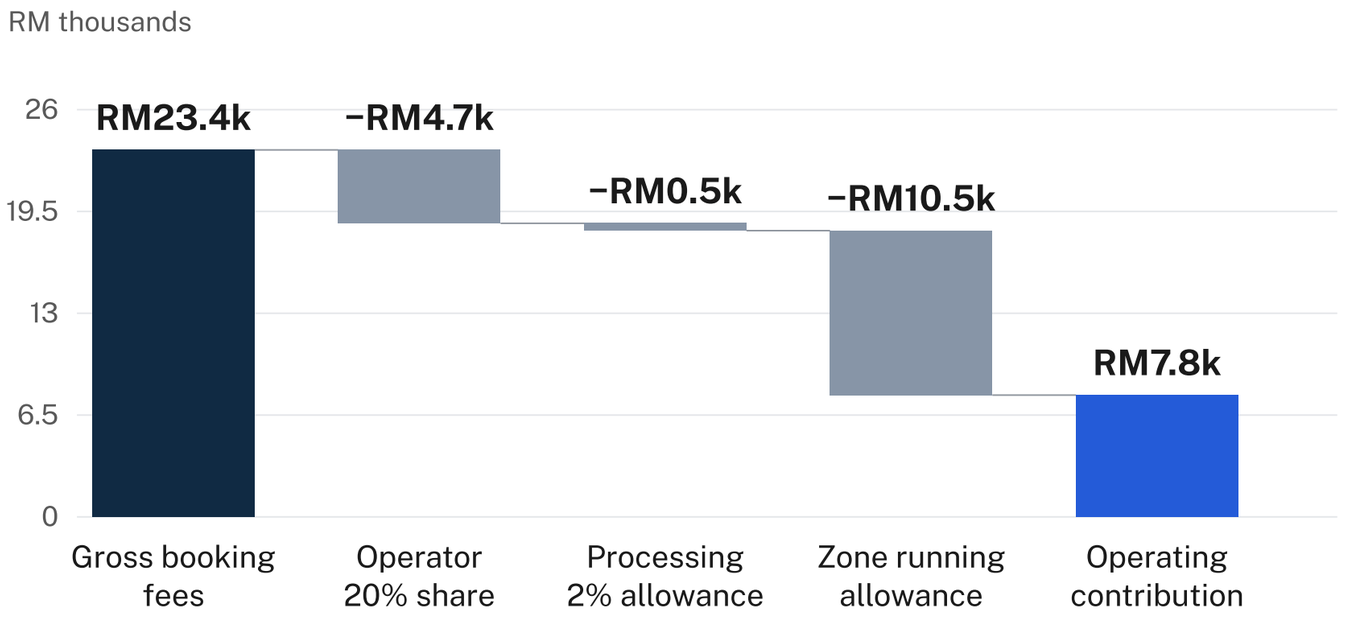
Procurement gate: no final installed-cost claim until scope, incumbent-system access and binding supplier quotes are reconciled.

Sources: M1: ParkFlow shared booking economics · S1: KL East Mall official parking information · S6: AAC current ANPR product listings · S7: Viskou DS-TVL224 display product. All amounts are planning allowances before unpriced scope, tax and contingency. Retailer listings do not establish installed cost.

Booking income contributes to the service

The model leaves RM7.8k a year before central support, tax and capital recovery.

Annual booking-fee bridge



365-day illustration; 104 peak / 261 weekday days.

Base planning inputs

Peak days	30 bookings / day × RM5
Weekdays	7.5 bookings / day × RM4
Allocations	20% operator + 2% processing
Zone running	RM10,500 per year

Demand, shares and costs require validation. Booking fees sit above normal parking charges.

Sources: M1: ParkFlow shared booking economics. Exact arithmetic: gross RM23,430.00 less RM4,686.00, RM468.60 and RM10,500 = RM7,775.40. Not fully loaded profit.

Three checks connect demand, cash and capital

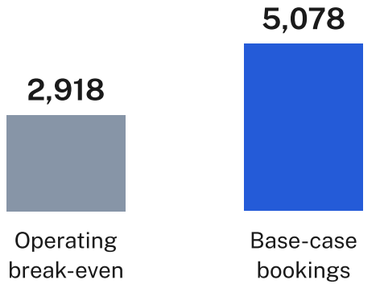
The operating case can be positive while the complete investment case still needs additional value.

One model across all three checks.

Fee mix → operating contribution → capital hurdle → incremental visit requirement.

Is zone OPEX covered?

Annual bookings, same fee mix

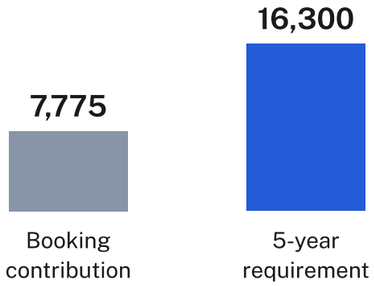


Base case clears zone OPEX. At the modeled fee mix, approximately 2,918 annual paid bookings cover RM10.5k. The base assumes 5,078.

Demand and payment costs still require validation.

Does booking repay capital?

Annual contribution (RM)

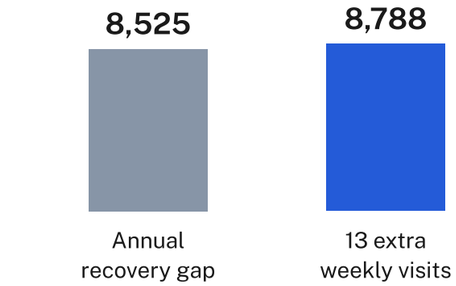


A recovery gap remains. RM7.8k booking contribution falls below the RM16.3k annual requirement for five-year simple recovery.

Additional net benefit required: RM8,524.60 a year.

Can visits close that gap?

Annual net benefit (RM)



Thirteen weekly visits illustrate the bridge. At RM260 spend and 5% effective net capture, 13 additional visits a week yield RM8,788 a year.

Capture and genuinely incremental visits remain pilot questions.

Sources: M1: ParkFlow shared booking economics · S4: Vase.ai, Malaysia Shopping Mall Report 2026 (n=500). Illustrative inputs, not a forecast. Excludes central support, tax and extra capital. Five-year recovery is a comparison case; 5% capture and RM260 basket are scenarios.

Test volume and support cost together

A positive zone contribution can disappear when central delivery effort is allocated.

Annual contribution before central support

Peak bookings / day	2.5 weekday / day	7.5 weekday / day	12.5 weekday / day
10	−RM4.4k	−RM0.3k	RM3.7k
20	−RM0.4k	RM3.7k	**RM7.8k · base**
30	RM3.7k	**RM7.8k · base**	RM11.8k
40	**RM7.8k · base**	RM11.8k	RM15.9k
50	RM11.8k	RM15.9k	RM20.0k

RM thousands; weekday / peak volumes are sensitivity inputs. Highlighted cell is the shared base case.

Support changes the outcome

RM0 support / year	**RM7,775**
RM12,000 support / year	**−RM4,225**
RM24,000 support / year	**−RM16,225**

Measure incidents, support minutes, refunds and actual payment fees before calling the service profitable.

Support allocations are comparison cases, not staffing estimates.

Demand test: track paid conversion, eligible-member exposure and repeat usage alongside the operating metrics.

Sources: M1: ParkFlow shared booking economics.

CAPItal recovery

The base case needs additional annual net benefits of RM8.5k for five-year simple recovery.

Annual contribution hurdle

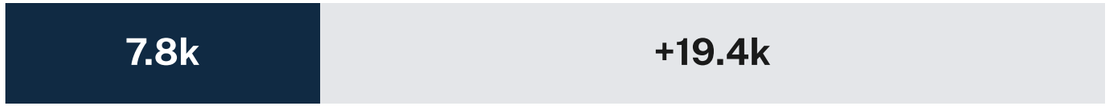
Annual net contribution required (RM thousands)

5-year simple recovery



Total RM16,300 / year

3-year simple recovery



Total RM27,167 / year

How to close the gap

- Verify genuinely incremental visits and the associated landlord capture.
- Quantify operational savings only after staff effort is measured.
- Treat sponsorship and external contracts as separate, signed upside.
- Include central support and any extra capital before testing the final hurdle.

RM81,500 initial capital; straight-line comparison, no discounting or terminal value.

Decision: agree SDP's actual hurdle rate and investment period before presenting an NPV, IRR or claimed payback.

Sources: M1: ParkFlow shared booking economics. Base booking-only simple recovery ≈10.5 years before excluded costs. Five/three years are comparison cases, not SDP-approved thresholds.

Retention sensitivity

The five-year recovery gap requires 7–29 additional weekly visits across the illustrated capture scenarios.

Weekly visits required to close the gap

Spend / visit	3% capture	5% capture	8% capture
RM195	29	17	11
RM260	22	13	8
RM327	17	11	7

Rounded up. Darker cells = more than 20 visits; blue-grey = 12 or fewer. Cells share the RM8,524.60 annual net benefit gap.

Worked case: RM8,524.60 ÷ (52 × RM260 × 5%) = 12.61, rounded up to **13 additional visits a week**.

Validate the conversion to value

- RM195 and RM327 are recalled survey baskets, not measured KL East baskets.
- RM260 is a midpoint illustration; 3%, 5% and 8% capture rates are scenarios.
- Effective capture must reflect actual leases and ancillary net contribution.
- Use incremental visits, avoiding double counting normal parking income or existing loyalty visits.

Formula: weekly visits = annual contribution gap ÷ (52 × spend per visit × effective mall capture).

Sources: M1: ParkFlow shared booking economics · S4: Vase.ai, Malaysia Shopping Mall Report 2026 (n=500). Capture is an effective net-to-mall scenario, not a verified GTO rental rate. Any incremental support cost raises the required visits.

Make ownership clear before deployment

One sponsor connects the operating, technology and investment decisions.

Accountable sponsor

SDP mall management

Owns the member proposition, pilot boundary and scale decision.



Parking operator	Admission, exceptions, staffing and ordinary parking settlement.	Decision evidence Operating agreement + signed SOP
Digital, IT & privacy	Interfaces, access, deployment and processing responsibilities.	Decision evidence API/SSO + data map + acceptance
Finance & procurement	Installed scope, support allocation, capture and return hurdle.	Decision evidence Comparable quotes + investment case

Team and integrator: co-design, software delivery and approved site connection; suppliers confirm warranty and handover.

Sources: S1: KL East Mall official parking information · S8: Malaysia PDP official Act/guidelines index. Accountabilities are proposed role assignments; no current contract, supplier selection or commitment is implied.

An eight-week trial with a baseline before launch

Separate service reliability from commercial uplift and define the comparison before the first paid booking.

Before live

Baseline & readiness

Survey the zone, confirm vendor access, agree member terms, label local frames and sign the measurement plan.

Weeks 1–2

Supervised service

Observe booked arrivals, reconcile counts and test plate mismatch, stale data and support handling.

Weeks 3–6

Controlled comparison

Use matched peak windows or approved blocked/randomized policy days. Track fulfilment and idle capacity.

Weeks 7–8

Value & investment review

Reconcile paid demand, support effort and cohort value. Extend measurement if the evidence is insufficient.

Operational comparison

Served accepted arrivals, reconciled count drift, idle held bay-hours while demand is refused, and the cause of every failure.

Commercial comparison

Exposed eligible members, paid conversion, matched/control revisit outcomes and complete support-adjusted cost.

Sources: M2: ParkFlow policy simulation, recorded output · M1: ParkFlow shared booking economics. Proposed design. Eligibility bias, seasonal demand, event mix and day-level clustering must be addressed in analysis; an eight-week trial is not guaranteed to prove retention.

Pilot scorecard

Agree the baseline, denominator and decision rule before the first paid booking.

Decision dimension	Metric / denominator	Proposed decision rule	Owner
Service reliability	Served accepted arrivals ÷ all accepted arrivals; report confidence interval.	Agree target and minimum observation volume; investigate every service failure.	Operator + mall ops
Demand	Paid bookings ÷ eligible exposed members; repeat usage by cohort.	Test RM4/RM5 willingness to pay; distinguish trial incentives from recurring demand.	Loyalty + finance
Capacity	Idle protected bay-hours while demand is refused; peak occupancy.	Improve against a valid comparator without reducing the service commitment.	Mall ops + analyst
Incremental value	Additional visits and net mall capture, adjusted for cohort differences.	Clear the full-cost hurdle with credible evidence; otherwise adapt or stop.	Finance + loyalty
Operating readiness	Manual interventions, outages, stale counts and unresolved data exceptions.	Signed SOP, fault drills, privacy review and auditable reconciliation.	IT + privacy + operator

These are proposed acceptance principles, not agreed numerical SLAs. An observation window must be extended if evidence is insufficient.

Design the exception path as carefully as entry

The first response to uncertainty is a safe operating state and a traceable recovery process.

Can the operator trust the current state?

Verified plate · current counts · live controller · valid entitlement

Yes

Apply the capacity guard

Admit eligible vehicles only when the agreed promise and buffer rules pass.

Owner: parking operator

No / uncertain

Pause new commitments

Use assisted verification, preserve egress, log the incident and reconcile before restarting.

Owner: operator + IT / integrator

Member exceptions

Published grace, no-show, overstay and refund rules.

Data incident

Restricted access, evidence preservation and SDP privacy escalation.

Low paid demand

Review scope and price with finance before expansion.

Sources: S8: Malaysia PDP official Act/guidelines index. Deployment obligations depend on the actual processing and operator arrangements; this is an implementation workstream, not a legal certification.

Scale the operating system in three horizons

The next horizon begins when evidence clears the gate, rather than when a date alone is reached.

Validated service	Repeatable operating system	Contracted platform
0–6 months	6–24 months	24–60 months
Prove at KL East	**Build the SDP system**	**Scale by contract**
• Survey + local baseline • Eight-week live trial • SOP + full-cost review	• Qualify Elmina + KLGCC • Shared interfaces + reporting • Repeatable support + onboarding	• External mall-group demand • Versioned APIs + SLA • Onboarding + support margin
Value gate: service and economics	**Repeatability gate: approved sites**	**Commercial gate: signed economics**



Three wholly owned malls

KL East, Elmina Lakeside and KLGCC. JV opportunities need separate rights, approvals and economics.

Sources: S5: SDP Q4 FY2025 quarterly report. Planning horizons and proposed milestones, not committed deployment or revenue forecasts.

Approve a bounded path to the pilot

Release deployment funding once the scope and measurement plan are confirmed.

Requested now

Authorize discovery. Set the pilot up to decide.

Nominate a KL East sponsor and operator counterpart. Enable the site survey, integration discussions and baseline measurement.

RM92k

Initial planning subtotal
Final installed scope pending

Return with four decision inputs

Feasible site	Zone boundary, incumbent interfaces and approved integration.
Complete cost	Installed quotes, support, tax and contingency.
Ready operator	Signed SOP, member terms and fault drills.
Credible value test	Baseline, cohort, measurement plan and go/adapt/stop criteria.

Sources: M1: ParkFlow shared booking economics. The deck requests a bounded pilot process; it does not imply an agreed contract, committed budget or proven commercial return.

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Thank you.

Team Hexa · Universiti Malaya

Questions & discussion



Supporting appendix follows.

Appendix / cost & scope

Keep planning allowances distinct from validated supplier prices and final installed scope.

Cost line	Allowance	Basis / validation status	Before commitment
Entry / exit ANPR	RM6,000	Legacy installed allowance; current AAC listings show multiple price labels.	Confirm reuse, exact model, warranty and installed quote.
Compact LED sign	RM4,000	Product option exists; the old RM2,033 price was not revalidated.	Confirm display size, viewing distance, interfaces and installed price.
Blind-spot cameras	RM4,000	Coverage-dependent allowance, not a counted site requirement.	Survey existing CCTV before adding cameras.
Cabling / electrical	RM7,500	Engineering allowance; installation distance not surveyed.	Quote PoE, conduit, network and power works.
Software integration	RM60,000	Prototype-to-production estimate; incumbent access not confirmed.	Define identity, payments, controller scope and acceptance.
Annual zone OPEX	RM10,500	Hosting, messaging and upkeep allowance.	Measure usage; allocate support, tax and contingency separately.

Not yet priced: zone barriers/boundary, controller/intercom changes, additional support, taxes, contingency and any extra privacy/IT deployment requirements.

Sources: S6: AAC current ANPR product listings · S7: Viskou DS-TVL224 display product · M1: ParkFlow shared booking economics.

Appendix / model reconciliation

Every sensitivity is derived from one fee, share, capital and operating-cost definition.

Pilot base model

Calculation	Annual RM
104 × 30 × RM5 + 261 × 7.5 × RM4	23,430.00
Operator share: 20% of booking fees	(4,686.00)
Processing: 2% of booking fees	(468.60)
Annual zone OPEX	(10,500.00)
Zone operating contribution	**7,775.40**
Capital required / 5 years	16,300.00
Additional net benefit required	**8,524.60**

What must not be mixed

- RM92k includes Year 1 running cost; RM81.5k is the upfront capital used in recovery comparisons.
- The old RM9,360 retention target is replaced with the model-derived RM8,524.60 five-year gap.
- Partner margin uses the same annual zone OPEX; cloud-only cost is not a complete contribution margin.
- No NPV/IRR from an older four-mall model is carried into this pilot case.
- Support, tax, contingency and additional capital remain explicit exclusions.

Sources: M1: ParkFlow shared booking economics.

Appendix / partner pace

Consistent zone OPEX lowers the illustrated partner contribution to RM29,250 per mall before central costs.

Sign-ups from Year 2	Year 7 live malls	Annual fees in Year 7	Annual zone OPEX	Contribution before central costs
0 / year	0	RM0	RM0	RM0
2 / year	12	RM540,000	RM189,000	RM351,000
4 / year	24	RM1,080,000	RM378,000	RM702,000
6 / year	36	RM1,620,000	RM567,000	RM1,053,000

Transparent assumptions

Six years of additions, 1.5 zones per partner, RM45,000 annual fee and RM10,500 annual running allowance per zone. Zero churn is assumed for this illustration.

Before using as an investment case

Verify signed pricing, churn, onboarding, support, hardware ownership, taxes and shared-platform allocation. These rows are annual run-rate scenarios, not cumulative cash or NPV.

Sources: M1: ParkFlow shared booking economics. RM45,000 − (1.5 × RM10,500) = RM29,250. No signed external-mall demand is represented.

Appendix / retention hurdles

Three-year recovery needs a larger incremental visit base than the five-year comparison.

Five-year gap: RM8,524.60 / year

Spend / visit	3% capture	5% capture	8% capture
RM195	29	17	11
RM260	22	13	8
RM327	17	11	7

Three-year gap: RM19,391.27 / year

Spend / visit	3% capture	5% capture	8% capture
RM195	64	39	24
RM260	48	29	18
RM327	39	23	15

Both tables: required additional weekly visits, rounded up; 52 weeks × basket × effective net capture. Add support costs before setting a final hurdle.

Sources: M1: ParkFlow shared booking economics · S4: Vase.ai, Malaysia Shopping Mall Report 2026 (n=500). RM195 / RM327 survey baskets; RM260 illustrative midpoint. 3–8% net capture scenarios are not verified SDP rent terms.

Appendix / technical validation

Back-testing, held-out-camera testing and synthetic policy trials answer different questions.

Experiment	Recorded evidence	Boundary	Next validation
Occupancy forecast	48,112 forecasts; six weeks; 30 Birmingham car parks; 60-minute MAE 2.07%.	Historical UK data; no KL East deployment result.	Local holdout by time, peak events and zone; calibrate intervals and drift.
Camera perception	78,270 patches; cameras 6–9 held out; accuracy 99.05%.	One outdoor Italian site; 540 occupied patches called free.	Indoor local labels, safety-critical false-free rates and count reconciliation.
Booking policy	1,000 synthetic busy days; forecast rule 100.0% baseline fulfilment.	Forecast-based future capacity differs from proposed hard current-capacity guard.	Replay deployed rule; validate queue, faults, no-shows and overstays.
Stress policy	Combined scenario: forecast policy 99.62%; fixed VIP 93.44%.	Scenario-specific assumptions; not “everything that can go wrong”.	Test wider conditions and field fault drills; report failures explicitly.

Sources: M2: ParkFlow policy simulation, recorded output · M3: ParkFlow ensemble back-test, recorded output · M4: ParkFlow camera benchmark, recorded output · S9: CNRPark dataset, CNR institute record · S10: UCI Parking Birmingham dataset.

Appendix / evidence & open gaps

Close documentary gaps through research; close site and commercial gaps through the pilot.

Claim / input	Evidence class	Treatment in this deck	Outstanding deliverable
KL East has LPR parking	Official public fact	Reuse-first integration approach.	Vendor permissions and zone-level interfaces.
Higher frequent-visitor basket	Survey association	Context for cohort selection, not causal uplift.	Local incremental visits and net capture.
Reliable, efficient policy	Internal simulation	Full comparison with explicit policy boundary.	Exact deployed-rule test and live fulfilment.
RM92k planning subtotal	Engineering estimate	Incomplete installed scope, not a final quote.	Supplier scope, support allocation and contingency.
Three-mall SDP rollout	Portfolio fact + proposed plan	Three wholly owned assets; each rollout gated.	Site feasibility, authority and business case.
External partner value	Illustrative economics	Consistent OPEX; no forecasted NPV claim.	Signed fees, demand, onboarding, churn and support.

Sources: S1: KL East Mall official parking information · S4: Vase.ai, Malaysia Shopping Mall Report 2026 (n=500) · S5: SDP Q4 FY2025 quarterly report · M1: ParkFlow shared booking economics · M2: ParkFlow policy simulation, recorded output.

Appendix / public sources

Primary sources support facts; they do not validate ParkFlow's untested commercial assumptions.

S1 · KL East Mall official parking information
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S2 · 1 Utama ONECARD official FAQ
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S3 · Sunway Pyramid official FAQ
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S4 · Vase.ai, Malaysia Shopping Mall Report 2026 (n=500)
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S5 · SDP Q4 FY2025 quarterly report
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S6 · AAC current ANPR product listings
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S7 · Viskou DS-TVL224 display product
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S8 · Malaysia PDP official Act/guidelines index
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S9 · CNRPark dataset, CNR institute record
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

S10 · UCI Parking Birmingham dataset
Public source checked 7 November 2026. Scope and limitations are recorded in the evidence register.

Full research notes: docs/deck-design-analysis/gaps-and-evidence.md. Former AAC 2025 PDF returned 404; legacy list prices were not promoted to live quotes.

Appendix / analysis & design record

The consulting edition is isolated from the original deck and uses a documented, measurable layout system.

Internal analysis files

- M1 · ParkFlow shared booking economics
- M2 · ParkFlow policy simulation, recorded output
- M3 · ParkFlow ensemble back-test, recorded output
- M4 · ParkFlow camera benchmark, recorded output

Recorded research outputs are not independent audits. Calculation and deployment boundaries remain visible throughout.

Design and evidence record

- guidance.md: measured grid, hierarchy, palette and density ramp.
- Gap register: corrections, sources and remaining site evidence.
- Reference PDFs: 45 pages measured; original deck: 32 slides measured.
- Opening remains lighter; business slides use comparative exhibits and explicit decision inputs.

Preservation: this edition has its own HTML, CSS, JavaScript and PDF. The current site-vnext deck and its assets remain unchanged.